

# Amirul Azizol

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## OVERVIEW

Software engineer based in Singapore with a foundation in research and software engineering.

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript/JavaScript, SQL, C++, Swift  
**Infrastructure & Tools:** AWS, GCP, Docker, Kubernetes, Jenkins, CI/CD, PostgreSQL, Git  
**Frameworks:** Langchain, Spring Boot, FastAPI, React, Next.js, Node.js, Angular  
**Machine Learning:** PyTorch, Transformers, Ray, PyArrow, LanceDB (RAG)

## EXPERIENCE

- Software Engineer** Jun. 2026 – Present  
*Dell Technologies*  
– DevOps and AgentOps for Dell's private cloud
- Machine Learning Engineer** Apr. 2025 – May. 2026  
*Apple (via TransPerfect)*  
– Delivered **300+** bugfixes and new features to enhance Malay Siri's intent detection, speech recognition, text-to-speech, and predictive keyboard.  
– Built synthetically generated datasets in low resource languages in collaboration with data curators.  
– Implemented regex-based filtering heuristics to reduce dataset noise, improving model accuracy by **23%**.  
– Developed an end-to-end testing tool, reducing manual QA effort by **80%**. (Python, Swift)  
– Refactored an internal data labelling tool with multithreading to achieve a **5x** speedup. (Python, C++)
- Software Engineer Intern** Jan. 2024 – Jun. 2024  
*Wavelet Solutions*  
– Optimized infrastructure and ensured reliability for an ERP processing **1M+** monthly financial transactions.  
– Engineered asynchronous ETL job processors to migrate **5M+** records to a centralized data lake. (SQL, AWS)  
– Designed RESTful APIs using Spring Boot, enforcing strict error handling to meet a **99.9%** uptime SLA. (Java)  
– Developed and deployed a UI module for 10+ applets, adopted by thousands of enterprise users. (Angular)  
– Refactored a mobile app to improve performance and cut page loading times by 30%. (Flutter)
- Embedded Software Engineer Intern** Nov. 2023 – Jan. 2024  
*Western Digital*  
– Engineered firmware features and failure analysis scripts for enterprise HDDs. (Python, C++)  
– Reduced the host interface team's bug backlog by **21%**.  
– Optimized CI/CD pipelines by improving retry logic, reducing false positive build failures. (Jenkins)

## PROJECTS

- Multimodal RAG for Medical and Forensic Analysis**  
– Top final year project in Summer '24 cohort.  
– Fine-tuned an 8B vision-language model, improving accuracy by **30%** over baseline by integrating RAG and CoT.  
– Architected a vector database with a custom chunking algorithm and a hybrid retrieval algorithm.
- RxNet**  
– Built a decentralized prescription tracking system with smart contracts to prevent fraud and improve traceability.  
– **Finalist** at APU Devmatch and **cash prize winner** at Monash Hackfest.
- Ren**  
– Developed a gamified environmental community app, ranked Top 10 at Kitahack (for Google Solution Challenge).

## EDUCATION

- Monash University** Feb. 2022 – Apr. 2025  
*Bachelor of Computer Science; Major in Advanced CS, Minor in Cybersecurity*  
– GPA: 3.9/4.0 | WAM: 87.1 (Top 1% of Cohort)  
– Recipient of two scholarships, Teaching Assistant for two units.